

Artificial Intelligence in Radiology

How fast it is growing, where it sits, and how much is pediatric

Pediatric Radiology AI project

June 13, 2026

Objectives

- Quantify how much **radiology AI** has grown, from a pre-deep-learning 2008 baseline to today, using reproducible queries against public databases.
- Measure **where** inside radiology the AI work sits — by imaging modality and by clinical task.
- Measure how much of radiology AI is **pediatric**, and in which modalities.
- Identify the **biggest players**: most-cited papers and most-used open-source software.
- Summarize what the field **does well**, what is **bleeding edge**, and what remains **unresolved** — for a children's hospital.

Databases queried:

- **PubMed** (NCBI E-utilities) — publication counts by year, by query.
- **OpenAlex** — citation counts and most-cited papers (union of modality/task searches, deduped by work id).
- **GitHub** — open-source tools ranked by stars.
- **DBLP** — ML/CV venue programs (CVPR, ICCV, NeurIPS, ICML, ICLR).
- **PatentsView** — granted US patents (requires an API key).

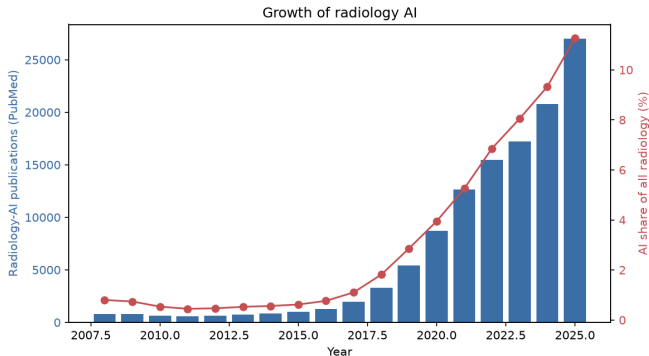
Representative PubMed query (radiology $\cap$ AI):

(radiology OR radiograph* OR “medical imaging” OR MRI OR CT OR ultrasound ...) **AND** (“artificial intelligence” OR “deep learning” OR “convolutional neural network” OR radiomics ...)

Counts use the [pdat] date facet; shares are ratios of counts. The exact strings live in `pedrad_ai/config.py`.

How much has AI grown in the radiology literature?

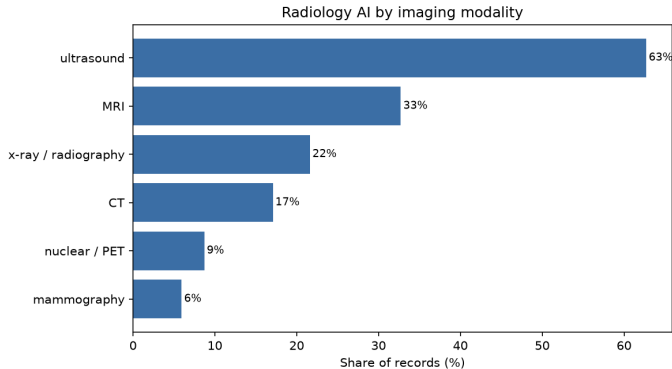
- 27,011 radiology-AI papers in 2025.
- Compound growth $\approx$ **23%/yr**.
- AI share of all radiology rose from **0.8%** (2008) to **11.3%** (2025).



Where the AI work sits — by modality

Share of radiology-AI papers mentioning each modality (overlapping):

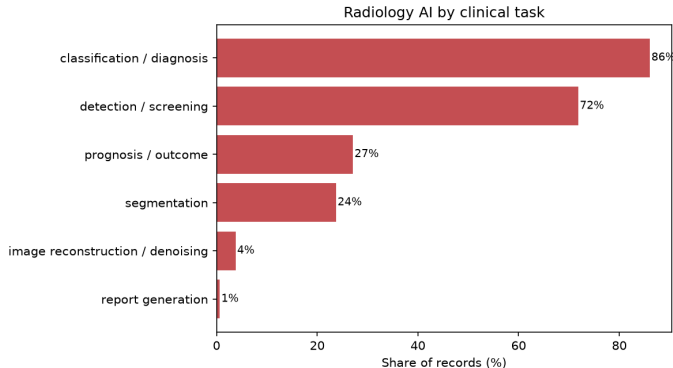
- ultrasound — 62.7%
- MRI — 32.7%
- x-ray / radiography — 21.6%
- CT — 17.1%



Where the AI work sits — by clinical task

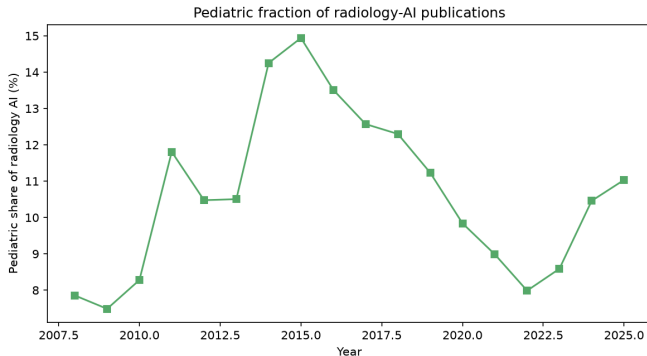
Share of radiology-AI papers by task (overlapping):

- classification / diagnosis — 86.1%
- detection / screening — 71.9%
- prognosis / outcome — 27.1%
- segmentation — 23.8%



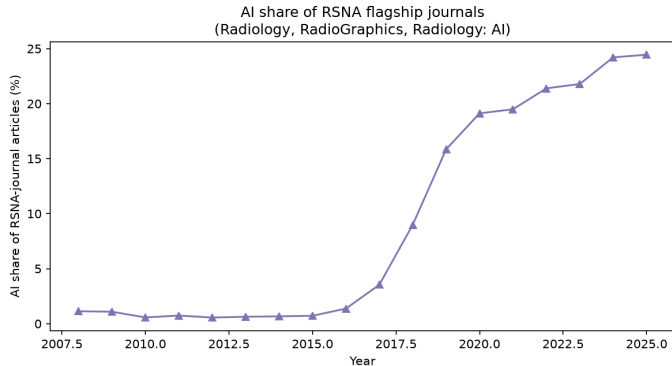
How much is pediatric?

- Pediatric work is **11.0%** of radiology AI in 2025 (2,980 papers).
- Small but growing; strongest in ultrasound and MRI.

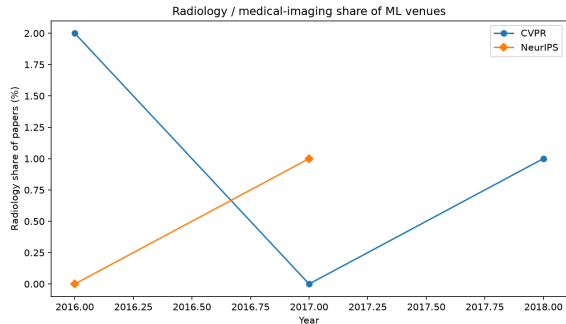


Radiology's own engagement: AI share of RSNA journals

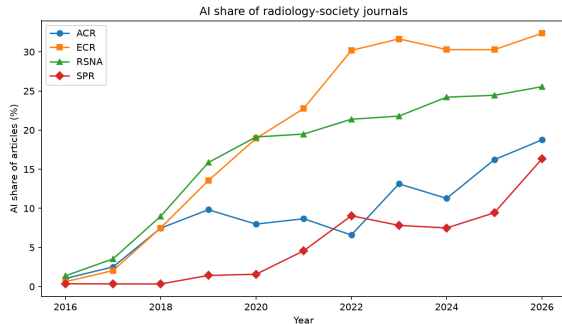
The AI share of RSNA's flagship journals rose from **1.1%** to **24.4%** — about one in four articles now touches AI.



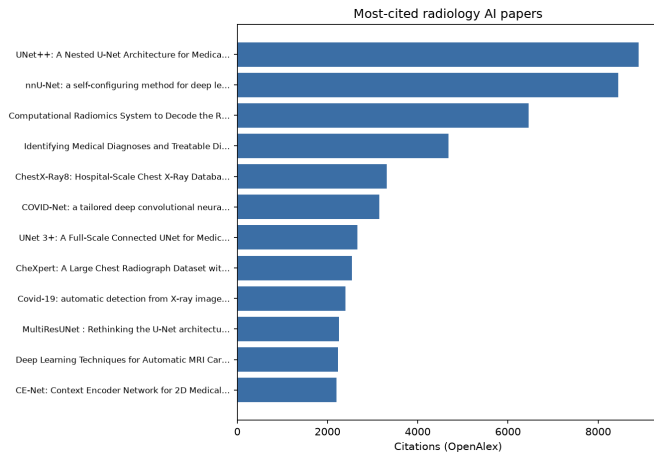
Radiology share of ML venues



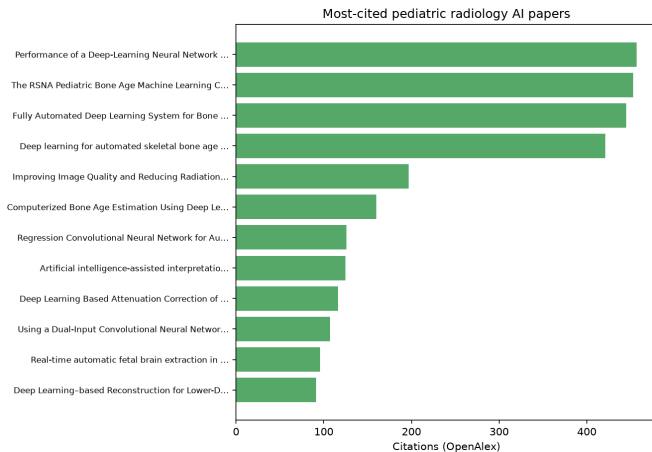
AI share of radiology-society journals



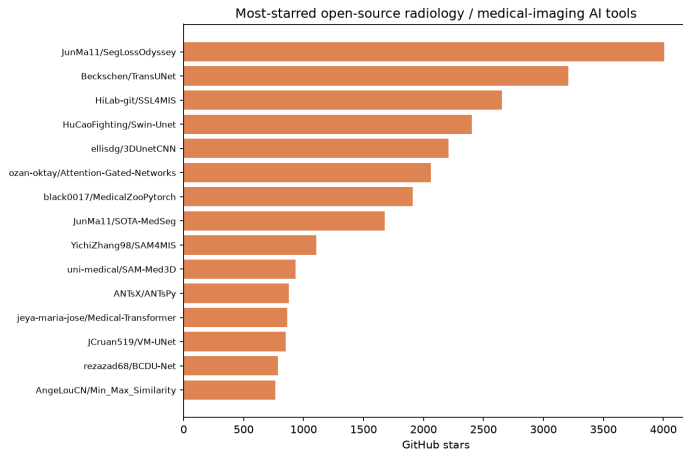
Biggest players — most-cited radiology AI papers



Biggest players — most-cited pediatric radiology AI papers



Biggest players — most-starred open-source tools



What radiology AI does well

- **Worklist triage:** hemorrhage, large-vessel occlusion, pulmonary embolism, pneumothorax — the most validated, most deployed category.
- **Detection / measurement aids** on high-volume adult exams (lung nodules, mammography, volumetry).
- **Image quality / acquisition:** deep-learning reconstruction and denoising — lower dose, shorter scans (most valuable in pediatrics).
- **Quantification / standardization:** segmentation, longitudinal tumor measurement.

- Foundation models and vision-language / report-generation systems.
- Opportunistic screening (bone density, coronary calcium, body composition).
- Multimodal and longitudinal models (imaging + EHR + priors).
- Self-supervised / label-efficient learning (key where labels are scarce).
- **Pediatric frontiers:** bone age (mature), fetal/neonatal brain MRI, congenital anomaly detection, scoliosis / Cobb-angle, growth-aware models.

What remains unresolved

- **Generalization:** models degrade across scanners, sites, populations.
- **Pediatric data scarcity & age dependence:** children are not small adults; adult models transfer poorly.
- **Prospective benefit:** most evidence is retrospective accuracy, not improved outcomes.
- **Report-generation trust:** LLM reports can be fluent and wrong.
- **Regulation/liability:** most cleared devices are validated on adults.
- **Workflow integration, monitoring, drift, and equity.**

Implications for a children's hospital

- ① **Buy maturity, build for the gaps:** adopt cleared adult-derived tools that transfer (triage, reconstruction); treat pediatric tasks as local validation / research.
- ② **Demand local pediatric validation** before clinical use.
- ③ **Prioritize dose and throughput:** deep-learning reconstruction gives the clearest pediatric benefit today.
- ④ **Plan for monitoring:** pediatric drift (growth, protocol change) is faster than in adults.

- Radiology AI grew from **0.8%** to **11.3%** of the radiology literature (2008–2025), $\approx 23\%/yr$.
- Pediatric work is **11.0%** of radiology AI — small but growing.
- RSNA journals: AI share **1.1%** $\rightarrow$ **24.4%**.
- Mature where data are large and adult (triage, reconstruction); pediatric bone age is the one mature pediatric task.
- Generalization, pediatric data, and prospective benefit remain the gaps.